Centrifugal Pump P-200 - Maintenance Manual (Rev C, 2023). The P-200 is a single-stage centrifugal pump for process water. Rated flow is 120 m3/h. Rated head is 45 m. The motor power is 22 kW. The maximum operating pressure is 10 bar. The maximum fluid temperature is 90 degC. Do not run the pump dry: dry running destroys the mechanical seal within minutes.

Routine maintenance. Lubricate the bearings every 2000 operating hours with lithium grease NLGI 2. Inspect the mechanical seal every 4000 operating hours. Replace the seal if leakage exceeds 10 drops per minute. The vibration limit measured at the bearing housing is 4.5 mm/s RMS. Record every maintenance action in the maintenance log of the plant.

Tightening torques. Casing bolts M12 must be tightened to 65 N.m. The impeller nut must be tightened to 120 N.m. Coupling guard bolts M8 must be tightened to 25 N.m. Always tighten bolts in a cross pattern, in two passes, using a calibrated torque wrench.

Troubleshooting. Low flow: check whether the suction strainer is clogged and whether air is entering the suction line. Bearing temperature above 80 degC: check for over-lubrication or shaft misalignment. Cavitation noise: the available NPSH is too low, the required NPSH of the P-200 is 3.5 m. Excessive vibration: check the alignment of pump and motor and the foundation bolts.